5. $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right| + c$

6. $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right| + c$

Example 2.40

Evaluate $\int \frac{dx}{16-x^2}$

Solution:

$$\int \frac{dx}{16-x^2} = \int \frac{dx}{4^2-x^2}$$
$$= \frac{1}{2(4)}\log\left|\frac{4+x}{4-x}\right| + c$$
$$= \frac{1}{8}\log\left|\frac{4+x}{4-x}\right| + c$$

Example 2.41

Evaluate $\int \frac{dx}{1-25x^2}$

Solution:

$$\int \frac{dx}{1-25x^2} = \frac{1}{25}\int \frac{dx}{\left(\frac{1}{5}\right)^2 - x^2}$$
$$= \frac{1}{25}\left[\frac{1}{2\left(\frac{1}{5}\right)}\log\left|\frac{\frac{1}{5}+x}{\frac{1}{5}-x}\right|\right] + c$$
$$= \frac{1}{10}\log\left|\frac{1+5x}{1-5x}\right| + c$$

DO YOU KNOW?

$$\int \frac{m}{a^2-(mx)^2}\,dx = \int \frac{d(mx)}{a^2-(mx)^2}$$
$$= \frac{1}{2a}\log\left|\frac{a+mx}{a-mx}\right| + c$$

Example 2.42

Evaluate $\int \frac{dx}{2+x-x^2}$

Solution:

By completing the squares

$$2+x-x^2 = 2-\left[x^2-x\right]$$
$$= 2-\left[\left(x-\frac{1}{2}\right)^2 - \frac{1}{4}\right]$$
$$= \left(\frac{3}{2}\right)^2 - \left(x-\frac{1}{2}\right)^2$$

$$\int \frac{dx}{2+x-x^2} = \int \frac{dx}{\left(\frac{3}{2}\right)^2 - \left(x-\frac{1}{2}\right)^2}$$
$$= \frac{1}{2\left(\frac{3}{2}\right)}\log\left|\frac{\frac{3}{2}+\left(x-\frac{1}{2}\right)}{\frac{3}{2}-\left(x-\frac{1}{2}\right)}\right| + c$$
$$= \frac{1}{3}\log\left|\frac{2+2x}{4-2x}\right| + c$$
$$= \frac{1}{3}\log\left|\frac{1+x}{2-x}\right| + c$$

Example 2.43

Evaluate $\int \frac{dx}{4x^2-1}$

Solution:

$$\int \frac{dx}{4x^2-1} = \int \frac{dx}{4\left(x^2-\frac{1}{4}\right)}$$
$$= \frac{1}{4}\int \frac{dx}{x^2-\left(\frac{1}{2}\right)^2}$$
$$= \frac{1}{4}\left[\frac{1}{2\left(\frac{1}{2}\right)}\log\left|\frac{x-\frac{1}{2}}{x+\frac{1}{2}}\right|\right] + c$$
$$= \frac{1}{4}\log\left|\frac{2x-1}{2x+1}\right| + c$$

DO YOU KNOW?

$$\int \frac{m}{(mx)^2-a^2}\,dx = \int \frac{d(mx)}{(mx)^2-a^2}$$
$$= \frac{1}{2a}\log\left|\frac{mx-a}{mx+a}\right| + c$$

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